

7. Biological control is used to reduce the numbers of a pest population through the introduction of another species. It has been used with varying success since the 19th century.

(a) (i) State **two** advantages of this method of control. [2]

Advantage 1

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Advantage 2

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(ii) State **two** disadvantages of this method of control. [2]

Disadvantage 1

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Disadvantage 2

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(b) The whitefly (*Trialeurodes vaporariorum*) is a pest which damages greenhouse crops such as tomatoes. Whitefly numbers can be reduced by using the biological control agent *Encarsia formosa*.

***Encarsia formosa* fact file**

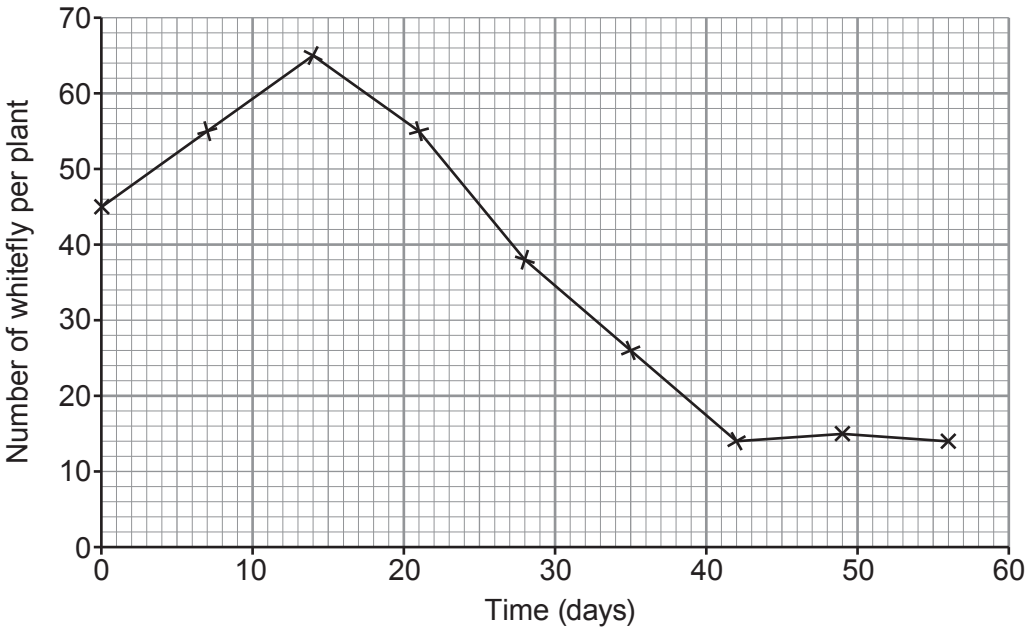
- *E. formosa* is a tiny wasp that lays eggs inside developing whitefly.
- When the eggs hatch, the young wasps kill the developing whitefly from the inside.
- Optimal conditions for *E. formosa* are temperatures over 20°C.
- When daytime temperatures are less than 17°C, *E. formosa* activity is significantly reduced, making it less effective.



Examiner
only

Graph 7.1 shows the number of whiteflies in a greenhouse containing tomato plants. *E. formosa* were introduced on day 7.

Graph 7.1



- (i) I. The use of *E. formosa* to reduce the number of whiteflies is considered to be successful when there are 20 or fewer whiteflies per plant. Use **Graph 7.1** to determine how many days it took for the number of whiteflies to fall to 20 following the introduction of *E. formosa*. [2]

..... days

- II. Suggest a reason why it took this long for the number to fall to 20. [1]

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- (ii) Suggest **one** reason why this method of pest control would not be effective to use if whiteflies damaged wheat crops grown in Wales. [1]

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- (iii) An alternative approach to reducing pest numbers is to use pesticide. State why it is not appropriate to use pesticide along with *E. formosa*. [1]

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